GATE CS/DA ASSESSMENT: ALGORITHMS & COMPLEXITY

Problem Sheet 2: Divide-and-Conquer Recurrence Relations

Question:
State the general form of the Master Theorem for divide-and-conquer recurrences. Using the Master Theorem, derive the exact tight asymptotic bound $\Theta(\cdot)$ for:

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n)$$